

Life history theory

switch back to introduce tradeoffs first?

Learning objectives

1. Know the terms and methods for analyzing structured populations
2. Convert life history information into a Leslie matrix
3. Get population growth rate from a Leslie matrix

Review of last lecture: environmental variability and how to deal with that modeled population growth with seed bank

assumed all individuals are same - in reality, different stages & ages within population

Population structure

Terminology

- demography - structured populations
- vital rates - describe each stage *germination, survival to reproduction, fecundity*
- age and stage *related but useful in different circumstances*
- stable stage distribution - proportion of individuals in each age/stage *even as total population size varies*

stage examples: seeds, vegetative, reproductive

Why care about population structure?

- more realistic *that births and deaths are age/stage structured*

example: seedlings die in frost, reproductive adults don't

- different stages, different pressures *seeds-soil conditions, adults-competition*
- compare individuals within stage

Modeling life history stages

let's say we have a plant with 3 ages: seed (0), seedling (not reproductive) (1), and adult (2)

seeds are in a seed bank

Leslie matrix

- quantify and analyze structured populations
- inputs: abundance, probability of transition to next stage, and fecundity
- calculate population growth rate and project abundance in each age

let's make an age-structured life table

Age	Description	Number Alive	Survival to Next Stage	Fecundity
0	Seed	1000	0.20	0

Age	Description	Number Alive	Survival to Next Stage	Fecundity
1	Seedling	200	0.25	0
2	Adult	50	—	100

let's put this in the leslie matrix format

- fecundity in first row
- each age is a column
- survival to next stage on the diagonal

```
leslie <- c(0,0.2,0,0,0,0.25,100,0,0)
matrix(leslie,nrow=3,ncol=3)
```

```
##      [,1] [,2] [,3]
## [1,]  0.0  0.00 100
## [2,]  0.2  0.00  0
## [3,]  0.0  0.25  0
```

can think of 0s as percentage of one stage going to others

for example, [3,1] is 0 because no seeds are going straight to adults

[2,2] is 0 because no seedlings are going back to seeds

Calculating λ

use linear algebra, not going to get into that, just some definitions

- eigenvalue = a scalar used in matrix multiplication
- several eigenvalues for a given matrix *i.e.* several scalars can work when doing this transformation/matrix multiplication
- largest, positive eigenvalue of a Leslie matrix *dominant - largest, positive* = λ = population growth rate

really easy to do in R, will do in lab

(maybe skip) Example of stage-structured population

- each column = stage *same as Leslie matrix*
- column 1 = rates from seeds to seeds/seedlings/adults
- column 2 = rates from seedlings to seeds/seedlings/adults
- adults to seeds is fecundity

```
stage <- c(0,0.2,0,0,0.1,0.3,50,0,0.8)
matrix(stage,nrow=3,ncol=3)
```

```
##      [,1] [,2] [,3]
## [1,]  0.0  0.0 50.0
## [2,]  0.2  0.1  0.0
## [3,]  0.0  0.3  0.8
```